

Stable Cu nanoclusters: from an aggregation-induced emission mechanism to biosensing and catalytic applications†

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Highly luminescent and stable Cu nanoclusters (NCs) have been prepared, displaying an intriguing aggregation-induced emission (AIE) feature. The attractive AIE feature allowed the CuNCs to serve as pH stimuli-responsive functional materials. Additionally, we explored the utility of CuNCs for biosensing and catalysis applications.

Metal nanoclusters (NCs) composed of a few to roughly a hundred of metal atoms bridge the gap between atoms and nanocrystals, and thus are of great fundamental interest as well as of potential practical importance.¹ Relative to their larger cousin metal nanocrystals, they possess sizes comparable to Fermi wavelength, and exhibit discrete electronic structure and molecule-like properties such as size dependent HOMO–LUMO transition and luminescence.² Among these metal NCs stabilized by various ligands or templates, thiolate-protected AuNCs and AgNCs are currently most intensively explored due to their attractive features such as chemical stability, ultrasmall hydrodynamic diameters, and tailorable surface properties. Particularly, they are very promising fluorescent probes for biomedical imaging and sensing owing to their favorable photophysical properties, ultrafine size, and low toxicity.³ Inspired by these intriguing properties, exciting progress in the development of fluorescent AuNCs and AgNCs has been achieved in the past decades.⁴ On the other hand, non-precious metal Cu is earth-abundant, significantly cheaper and widely used in industries. Theoretical and experimental studies have shown that CuNCs possess unique catalysis and photoluminescence (PL) properties.⁵ Nevertheless, due to the synthetic difficulty in controlling ultrafine size and the susceptibility to oxidation upon exposure to air, it is a significant challenge to prepare stable CuNCs, and thereby studies on CuNCs are still in a preliminary stage relative to what has been achieved for both Ag and Au.⁶ There are only limited reports in the literature on the synthesis of thiolate-protected CuNCs.⁷ Recently, Chen's group developed one-pot synthesis of CuNCs based on the

modified Brust–Schiffrin method.^{5a} However, the as-synthesized CuNCs are organically soluble only with low quantum yield (QY) and poor stability. Developing a straightforward and effective method to synthesize brightly luminescent and stable CuNCs is highly desirable, which, if realized, would open up possibilities of an array of their practical application.

Fundamental understanding of the emission mechanisms would lead to a better synthesis route for strongly luminescent CuNCs. However, the origin of PL is a fundamental issue which is still not well addressed. Typically, the PL may derive from the metal core (quantum size effect) and from surface metal (the interaction between the metal core and surface ligands).⁸ Whereas the debate from both sides still needs further detailed analysis and experiments. Recently, we discovered that CuNCs generate a striking fluorescence intensity enhancement upon solvent induced aggregation.⁹ Although the promise of their application in biosensing is hindered by the relatively low QY and aggregation-induced emission enhancement in non-aqueous solution, it provides a clue for the synthesis of highly luminescent CuNCs.

Herein, we develop a facile, one-pot, “green” synthesis of intensely luminescent and stable CuNCs with QY as high as 16.6%. Intriguingly, the as-prepared CuNCs exhibit aggregation-induced emission (AIE) behavior: molecules non-emissive in the solution state are induced to emit light by the formation of aggregates. The attractive AIE feature allowed the CuNCs to serve as pH stimuli-responsive functional materials. Facilitated by bright luminescence and good stability, we explored the utility of CuNCs for biosensing and catalysis applications.

In a typical experiment, a 100 mM Cu(NO₃)₂ aqueous solution (40 μL) was added into a fresh aqueous solution (4 mL) containing 10 mM D-penicillamine (DPA) under vigorous stirring at room temperature. As shown in Fig. S1 (ESI†), the evolution of the absorption and emission spectra of CuNCs indicated that the strongest PL was obtained under stirring for 90 min. The resultant CuNCs were poorly soluble under the reaction conditions (pH 2.66) and spontaneously precipitated out of solution. A powder sample was collected for analyzing the CuNCs in detail. The tentative suggestions for the formation of CuNCs are provided in the ESI† (Fig. S2 and S3).

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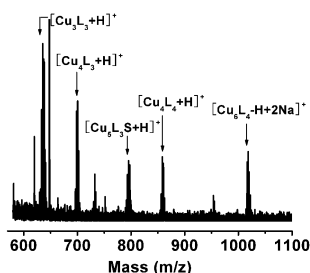


Fig. 1 Positive mode MALDI-TOF MS of CuNCs.

To determine the exact number of Cu atoms and ligands of single CuNCs, mass spectrometric studies were performed. Fig. 1 shows the representative matrix-assisted laser-desorption ionization time of flight mass spectrum (MALDI-TOF MS) of CuNCs, which was acquired in the positive-ion mode. Facilitated by the isotopic mass distribution of Cu, the highest m/z peak is assigned to $[\text{Cu}_6\text{L}_6\text{-H} + 2\text{Na}]^+$ (where $\text{L} = \text{C}_5\text{H}_{10}\text{O}_2\text{NS}$), existing in the form of Na adducts. As shown in Fig. S4 (ESI[†]), the base peak at 1018.14 is in perfect agreement with the theoretical exact mass of 1018.71 (deviation 0.57). And the observed isotope pattern matches perfectly with the theoretical prediction. The lower mass peaks are ascribed to $[\text{Cu}_4\text{L}_4 + \text{H}]^+$, $[\text{Cu}_5\text{L}_5 + \text{H}]^+$, $[\text{Cu}_4\text{L}_3 + \text{H}]^+$, and $[\text{Cu}_3\text{L}_3 + \text{H}]^+$, respectively. Further confirmation came from electrospray ionization mass spectrometry measurements, which showed characteristic signatures of CuNCs (Fig. S5, ESI[†]). A series of m/z peaks are assigned to $[\text{Cu}_6\text{L}_6\text{-H}]^-$, $[\text{Cu}_5\text{L}_4\text{-H}]^-$, $[\text{Cu}_4\text{L}_4\text{-H}]^-$, $[\text{Cu}_4\text{L}_3\text{-H}]^-$ and $[\text{Cu}_3\text{L}_3\text{-H}]^-$, respectively, consistent with the result of the MALDI-TOF MS. The FTIR spectra were exploited to gain further insight into chemical and surface properties of the prepared CuNCs (Fig. S6, ESI[†]). The disappearance of the S-H stretch mode at 2527 cm^{-1} suggested that DPA molecules anchored on the surface of CuNCs through the Cu-S bonding. The oxidation state of the CuNCs was determined by X-ray photoelectron spectroscopy and Auger electron spectroscopy (Fig. S7, ESI[†]), indicating that the CuNCs are composed of Cu^+ . This is consistent with the mild reduction environment created by the weak reducing agent DPA and such small clusters (3–6 atoms).

Fig. S8 (ESI[†]) illustrates the absorption spectrum of an aqueous CuNC solution. A strong, featureless absorption band in the UV region decays roughly exponentially toward the visible region. Upon excitation at 345 nm, the CuNCs displayed two distinct emission bands: a weak one centered at 418 nm and a strong one at 640 nm (Fig. 2A). Strikingly, the total QY of CuNCs in aqueous solution at room temperature was determined to be as high as 14.1%. After lyophilization, the QY of the solid CuNCs was found to be slightly higher,

16.6%. Bright red emission was observed when aqueous solution of CuNCs was deposited on the surface of glass, silicon and cotton owing to the formation of aggregates (Fig. S9, ESI[†]). When used as luminescent ink on the commercially available paper, the written words are invisible in daylight; but interestingly, red luminescent “copper nanoclusters” are observed when subjected to UV radiation. In sharp contrast, the conventional luminophores suffer from the notorious effect of aggregation-caused quenching in the condensed phase.¹⁰ It is anticipated that the unique AIE characteristic can lay a foundation for the potential applications of the as-synthesized CuNCs in anti-counterfeiting, optoelectronics and sensory systems.¹¹ Fig. S10 (ESI[†]) shows that these CuNCs display good photostability. Notably, the as-synthesized CuNCs are stable over one year of storage in the solid state by storing in the refrigerator without the protection of inert gas. To our knowledge, this was the first successful attempt on “green” synthesis of intensely emissive and stable CuNCs.

To get insight into the PL mechanism, we measured the PL lifetimes of the CuNCs at different emission wavelengths using a time-correlated single photon counting technique (Fig. S11 and Table S1, ESI[†]). The average lifetime of emission at 425 nm is 1.78 ns, attributed to emission from singlet excited states, which has been previously observed in the PL decays of many fluorescent AuNCs.¹² However, the average lifetimes at 600 nm and 645 nm were calculated to be much longer, 139.0 μs and 150.6 μs , respectively. The long excited-state lifetime in the hundreds of microsecond range and the large Stokes shift (295 nm) suggested that the red emission of the CuNCs could be tentatively assigned to triplet excited states. Namely, the ligand-to-metal charge transfer (LMCT) character mixes with metal-centered states which are modified by Cu-Cu interactions, which is consistent with the observed charge transfer behavior between sulfur atoms of DPA and $\text{Cu}(\text{I})$.^{5a,8,9,13} Only one main PL band at 640 nm which was essentially invariant with the size of clusters also demonstrated that the emission likely arises from the surface state.¹⁴ Because of a high fraction of $\text{Cu}(\text{I})$ species on the cluster surface, the emission at 640 nm is much stronger than that at 418 nm, which is in contrast to that in the previous report.^{5a}

As shown in the Fig. 2B inset, the turbid CuNC suspension emitted an intense red light under UV light. In striking contrast, the clear one obtained by adjusting pH to 7.0 displayed no PL. As a variation in the pH of an aqueous medium changes the structure of CuNCs from aggregates to isolated species, it is anticipated that the CuNCs can work as luminescent pH chemosensors. The PL intensity decreased linearly with the pH from 5.0 to 6.2, which is the most sensitive range for intensity response (Fig. 2B). At a pH of 6.2, the emission is approximately 100-fold lower than that at pH 4.0. To evaluate the reversibility of the switching operation upon variation of pH, the as-prepared CuNCs were subjected to pH cycling between pH 3.1 and 7.1 using acid and base as modulators. As shown in Fig. 2C, the luminescence switching operation can be repeated for five consecutive cycles without fatigue, indicative of good reversibility of the two-way switching processes. In the aqueous solution with high pH values, the CuNCs exist in the deprotonated form (COO^-) and are dissolved as isolated species. The clear one displayed no PL, partially due to PL quenching by intramolecular motions during the long lifetime of the triplet state.¹⁵ The intramolecular motions serve as non-radiative channels for its excited states to decay. Upon decreasing the pH value, CuNCs were

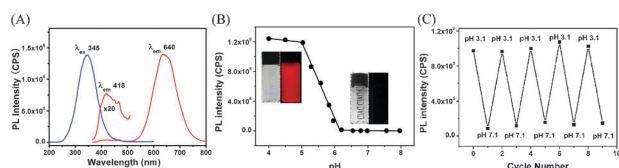


Fig. 2 (A) PL spectra of the aqueous CuNC aggregates (pH 2.66); (B) plots of PL intensity of CuNCs against pH; inset: photographs of the CuNCs in the aggregation and the dispersed state under the irradiation of (left) visible and (right) UV light; (C) PL intensity upon the cyclic switching of the CuNCs under alternating conditions of pH 3.1 and pH 7.1.

transformed to a neutral form. Then the CuNCs were aggregated through noncovalent molecular interactions, such as hydrogen bonds between the carboxyl groups, which are involved to greatly stiffen molecular conformation and arrangements, which reduces the energy loss through the nonradiative rotational relaxation channel and thus strikingly enhances the emission efficiency.^{15,16} Also we studied the PL behaviour of CuNCs by a solvent-induced aggregation method. As illustrated in Fig. S12 (ESI[†]), addition of large amounts of ethanol causes the CuNCs to aggregate and induces them to emit efficiently. Scanning electron microscopy and transmission electron microscopy (TEM) were used to probe the morphology of aggregates. As shown in Fig. S13 (ESI[†]), the CuNCs self-assemble into a lamellar structure and the lamellar structure further stack to a multilayer structure. Upon addition of NaOH (pH 7.0), the aggregates were dissolved. Restricted by the resolution of TEM, the tiny CuNCs could not be observed. Only tiny aggregates of several CuNCs with an average diameter of 1.0 ± 0.4 nm are discernible in the HRTEM image (Fig. S14, ESI[†]).

Owing to their bright luminescence and stability, the as-prepared CuNCs should have promising applications in biosensing and catalysis. H_2O_2 is an important intermediate species in many biological processes, which is generated upon the catalytic oxidation of the respective substrates by O_2 -dependent oxidase. The PL of the CuNCs is effectively quenched by H_2O_2 . During the course of reaction, the Cu centers were oxidized to Cu^{2+} , which was supported by XPS data (Fig. S15, ESI[†]). The PL spectra of the CuNCs in the presence of variable concentrations of H_2O_2 are shown in Fig. 3A. A linear range is observed from 0.05 to 2 mM (Fig. 3A, inset), and a limit of detection (LOD) of 0.01 mM for H_2O_2 analysis is achieved.

The effective quenching of CuNCs by H_2O_2 facilitates the probing of an oxidase-stimulated H_2O_2 -generating process. This is exemplified via the glucose oxidase-stimulated oxidation of glucose through the enzyme-catalyzed formation of H_2O_2 . As depicted in Fig. S16 (ESI[†]), PL intensity of CuNCs is decreased proportionately with an increased concentration of glucose. A linear relationship was observed between the PL intensity ratio (F_0/F) and the concentration of added glucose from 0.01 mM to 0.8 mM, and the LOD for glucose that was determined to be 0.01 mM. Moreover, there was negligible influence upon the addition of glucose analogues, such as fructose, lactose, sucrose, and maltose. The results allowed CuNCs to serve as luminescent probes to follow the activity of other oxidases.

Due to the ultrafine size, the CuNCs are expected to exhibit significant catalytic performance.^{5c} The catalytic activity was examined

by using the reduction reaction of methylene blue (MB) by hydrazine (N_2H_4) as model reaction. As shown in Fig. 3B, the original absorption peak of MB- N_2H_4 solution is at 654 nm, and the absorbance intensity remains almost unaltered. Addition of CuNCs to the reaction mixture causes the bright blue color of the solution to completely vanish within 1 min. The reaction rate constant k_0 was determined to be $(8.6 \pm 0.8) \times 10^3 \text{ M}^{-2} \text{ s}^{-1}$, suggesting that CuNCs serve as effective catalysts in the reaction (Fig. S17, ESI[†]). The high catalytic activity of CuNCs is attributed to that the LUMOs of CuNCs are located above the MB redox potential and below the N_2H_4 redox potential.^{5c} In addition, the CuNCs retain their catalytic activity for 7 cycles (Fig. S18, ESI[†]), suggestive of good recyclability in the catalytic reaction.

To explore the generality of the synthesis strategy, we further extended to prepare DPA-mediated AgNCs. PL spectra and photographs of AgNCs are shown in Fig. S19 (ESI[†]).

In summary, we presented a facile method for the preparation of highly luminescent and stable CuNCs. Significantly, the CuNCs feature unique AIE characteristics, and serve as pH stimuli-responsive functional materials. Facilitated by bright PL and good stability, the CuNCs exhibited excellent performance when serving as an effective catalyst and a sensitive probe for biosensing. In view of the convenient synthesis route and attractive luminescent properties, the synthesized metal (Ag, Cu) NCs are quite promising in applications such as imaging, sensing, catalysis and photovoltaics.

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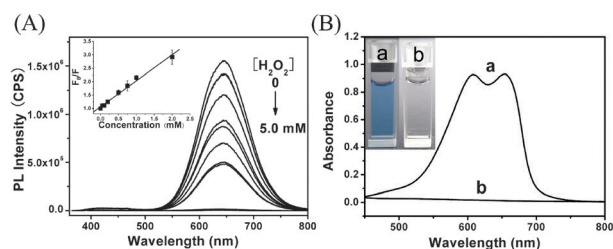


Fig. 3 (A) Effect of H_2O_2 concentration on the PL intensity of the CuNCs; inset: plot of the PL intensity ratio F_0/F against the H_2O_2 concentration over the range 0–2.0 mM. (B) UV-vis spectra of (a) the initial mixture with the MB dye and N_2H_4 , and (b) the mixture reacting for 2 min after the addition of 100 μM CuNCs; inset: photographs of solutions (a) and (b).